



Autonomous Drone Tutorial

Flapper Drones

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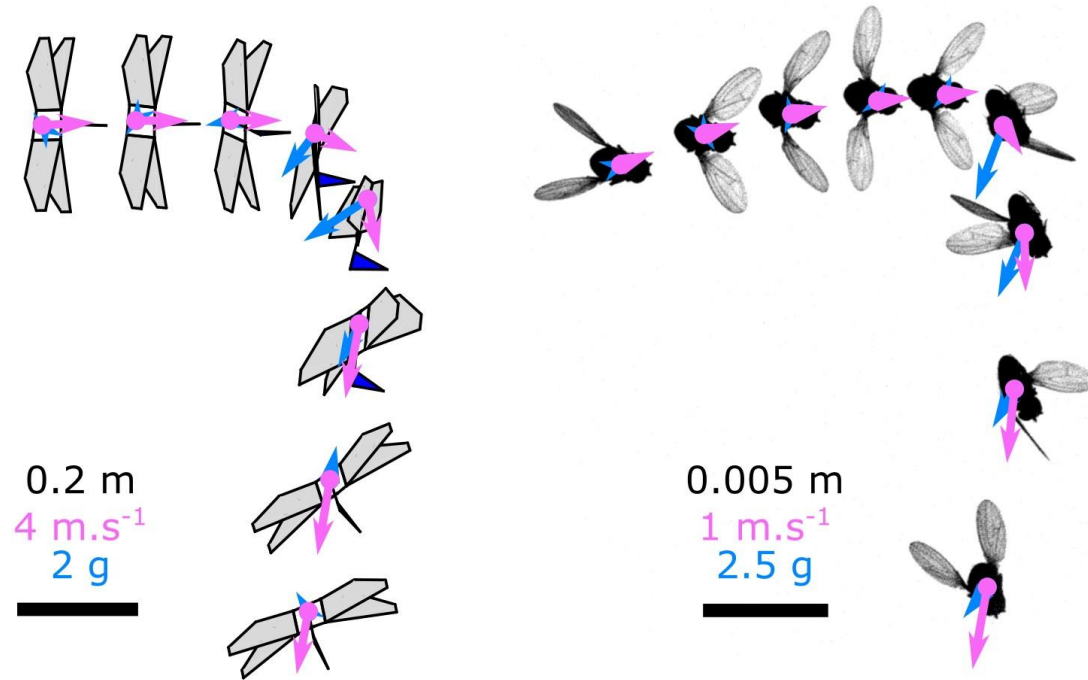
About Flapper Drones

- Founded in 2019 in the Netherlands (Spinoff TU Delft)
- Since 2022 in Jablonec nad Nisou, Czechia
- Bioinspired R&D platforms
- Indoor drone performances (collaboration with Wombatronic)



Research spinoff

Evasive maneuvers of fruit flies



Flapper Nimble+



SPECIFICATIONS:

Min. takeoff weight:	102 g
Max. payload:	30 g
Wingspan:	49 cm
Flapping frequency:	13-15 Hz (hover) 20 Hz (max)
Actuators:	2 x BL-DC 2 x rotary servo
Battery:	300 mAh 2S LiPo
Flight time:	8 minutes 5 minutes (hover with max. payload) Up to 14 minutes with larger batteries
Open-source firmware	
Open-source Python API	
Indoor Positioning Systems	



Flight control hardware & optional sensors

Crazyflie Bolt

STM32F4 MCU

- I2C, SPI, UART
- 4 x GPIO

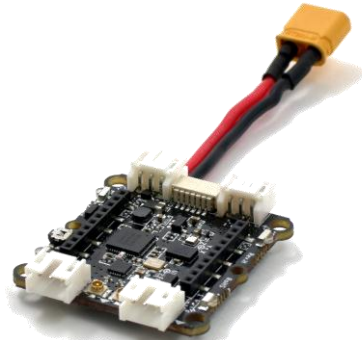
Bosch BMI088 IMU

nRF51822 SoC 2.4 GHz radio

External receiver

WS2812b RGB (3.8V, 1A)

microSD (data logging)



Crazyflie sensor decks

Multi-laser Ranger (ToF)

Flow deck (optic flow)*

AI deck* (camera + processing)

UWB deck*

** Experimental support*



Indoor Positioning Systems

HTC/Steam VR Basestations

(via Crazyflie Lighthouse deck)

Optical Motion Capture

(OptiTrack, Vicon, Qualysis, ...)

UWB Radio Beacons*

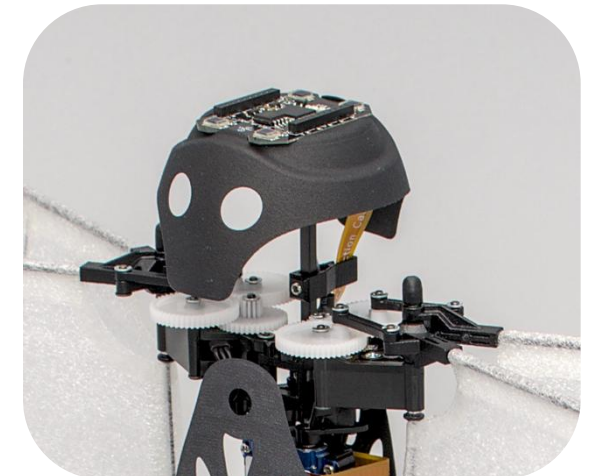
(via Crazyflie Loco Positioning System)



Directly mounted deck



Extension cables



Integration into body shells

Tutorial setup

Crazyflie Bolt

STM32F4 MCU

- I2C, SPI, UART
- 4 x GPIO

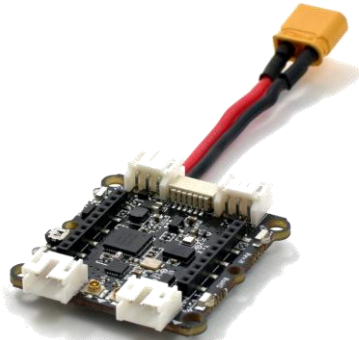
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Crazyflie SW Ecosystem

Open-source firmware

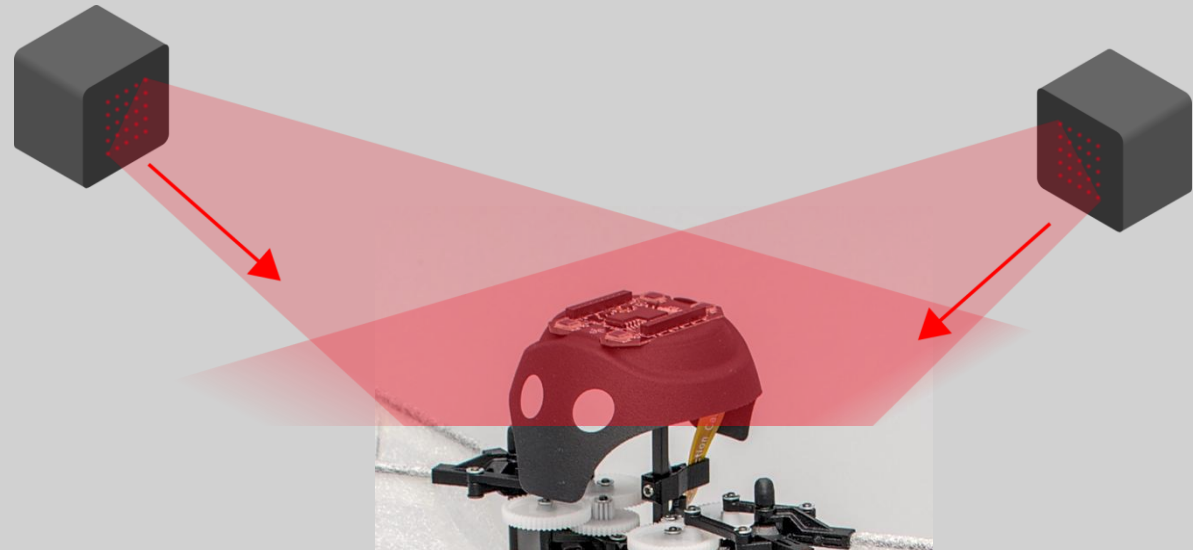
<https://github.com/bitcraze/crazyflie-firmware>

Open-source Python API

- High level control
- Swarming

<https://github.com/bitcraze/crazyflie-lib-python>

Lighthouse tracking system



Tutorial setup

- Geofence
 - The Flapper *should* not be able to fly out of a virtual cage
- LED status (top LEDs)
 - Color – battery level
 - Blinking – geofence activated (setpoint outside)
- Autoland
 - Flappers should autoland when low battery
- URI – *Unique Resource Identifier*
 - [DongleID]/[Radio Channel]/[Address]

Connecting to Raspberry Pi

- 1) Download **TigerVNC** (or see shared folder for TigerVNC files)
- 2) Connect to „**Flapper WiFi**“
Password: „NimblePlus“
- 3) Run Tiger VNC and connect to the following VNC server:
Group 1: „192.168.0.201“
Group 2: „192.168.0.202“
- 4) Confirm certificate exception (2x prompt)
- 5) Use these credentials:
Username: „pi“
Password: „password“



Tutorial example scripts

- Example 0:
 - Python basics
- Example 1:
 - Connecting to the Flapper
 - Controlling flight using the position HL commander: left, right, up...
- Example 2:
 - Controlling flight using the High Level Commander: goto, yaw, spiral
 - Setting drone parameters (RGB Lights color)
- Example 3:
 - Logging data while flying and plotting it
 - Swarming class allowing to interact with multiple Flappers/Crazyflies
 - Using live sensor data of a hand-held Crazyflie (Magic Wand) to control a Flapper

Example 1

- Run the script, link what you observe with the code
- Adjust the script to make the Flapper fly to your liking

Example 2

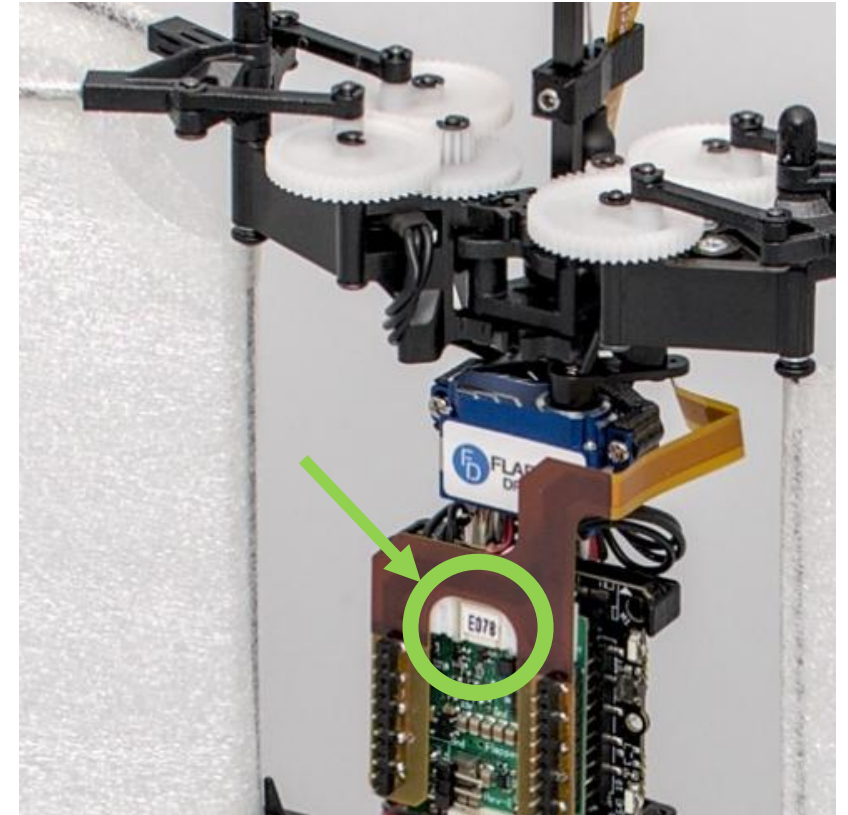
- Run the script, link what you observe with the code
- Adjust the script to make the Flapper fly to your liking
 - Did it work the way you wanted?
 - Iterate to get the desired flight trajectory
 - Add RGB light effects
 - Can you design a trajectory around a human standing in the center?

Example 3

- Run the script, link what you observe with the code
- Adjust the script to make the Flapper respond to the „Magic Wand“ movements as you would like
 - Can you amplify the Magic Wand motion? (small wand movements → larger Flapper movements)
 - Can you also control the yaw angle?
 - Could the magic wand roll/pitch angles be used to steer the Flapper's velocity?

URI

- URI – *Unique Resource Identifier*
 - radio://[DongleID]/[Radio Channel]/[Address]
 - Address: „FD##“
where ## are the last 2 digits of the Flapper's ID
 - Group 1: Radio Channel 04
 - Group 2: Radio Channel 08



Useful resources

- Crazyflie Python Library documentation:

<https://www.bitcraze.io/documentation/repository/crazyflie-lib-python/master/>

- Crazyflie Python Library source code:

<https://github.com/bitcraze/crazyflie-lib-python/tree/master/cflib>

- Flapper Nimble+ documentation

<https://docs.flapper-drones.eu/doku.php?id=nimbleplus:specs>